Vampire-like BB(6) machines

Bell map driven by the digits of one large number and a short word or tail; a HALT needs a certain sequence of residues (“lottery”). Expected values: Markov estimate with random residues (heuristic, not a proof). Green = HALT found, yellow = open. Status of 14 Nov 2026.

machine	type	status	expected HALT (Markov)	computing time to HALT	steps of the original TM

🟢 `1RB0RB_1LC1RF`
`0RD0LB_0RA1LE`
`1LD---_0RB0RC`
	vampire, 4/3, closed table (192 words)	**simulated to HALT + proof** (Bell 298,793,335, 21 Oct; Rocq proof busycoq PR #41, accepted)	$1.6 \cdot 10^{-9}$ per Bell → ≈ $6 \cdot 10^{8}$ Bells	≈ 2 min on one core (belldc3)	≈ $10^{(7.5 \cdot 10^{7})}$

🟢 `1RB0RF_1LC---`
`1RD0LC_0RE0RD`
`1LF1LC_1LE1RA`
TM1-F, orbit O2
	vampire family, 3/2, closed table (2,880 words)	**simulated to HALT** (Bell 731,899,010,932, Ahmer Raza, 30 Oct; confirmed by Thomas Reus)	$8.9 \cdot 10^{-13}$ per Bell → ≈ $1.1 \cdot 10^{12}$ Bells (median $7.8 \cdot 10^{11}$)	23 h 42 min on 8 cores (AMD EPYC 9655, belldco); ≈ 13–14 days on a notebook	≈ $10^{(2.58 \cdot 10^{11})}$

🟢 `1RB---_1LC0RB`
`0LD0LC_1RE1RB`
`1RD1LF_1LA0LE`
TM1, orbit O1
	as TM1-F	**simulated to HALT** (Bell 747,242,483,684, Ahmer Raza, 30 Oct)	as TM1-F	24 h 29 min on 8 cores	≈ $10^{(2.63 \cdot 10^{11})}$

🟢 `1RB0LA_0RC0RB`
`1LD1LA_1LC1RE`
`1RF0RD_1LC---`
TM2, orbit O2
	as TM1-F	**simulated to HALT** (orbit O2, as TM1-F)	as TM1-F	as TM1-F	≈ $10^{(2.58 \cdot 10^{11})}$

🟢 `1RB0RE_1RC---`
`0RD0RC_1LE1LF`
`1LD1RA_1RC0LF`
TM2', orbit O2
	as TM1-F	**simulated to HALT** (orbit O2, as TM1-F)	as TM1-F	as TM1-F	≈ $10^{(2.58 \cdot 10^{11})}$

🟢 `1RB0LA_0RC0RB`
`1LD1LA_1LC1RE`
`1RF0RD_1LA---`
TM2a, orbit O3
	as TM1-F	**simulated to HALT** (Bell 1,961,935,897,288, Ahmer Raza, 30 Oct)	as TM1-F	79 h on 8 cores (65 h of new progress)	≈ $10^{(6.91 \cdot 10^{11})}$

🟢 `1RB---_0RC0RB`
`1LD1LE_1LC1RF`
`1RB0LE_1RA0RD`
TM2b, orbit O4
	as TM1-F	**simulated to HALT** (Bell 1,941,376,761,248, Ahmer Raza, 1 Nov)	as TM1-F	several days on 8 cores	≈ $10^{(6.84 \cdot 10^{11})}$

🟢 `1RB0RF_1LC---`
`1LF1LD_1RE0LD`
`0RC0RE_1LC1RA`
TM2b', orbit O4
	as TM1-F	**simulated to HALT** (orbit O4, as TM2b)	as TM1-F	as TM2b	≈ $10^{(6.84 \cdot 10^{11})}$

🟡 `1RB1LF_1LC0RD`
`---1LA_0LA0RE`
`1LA1RA_1LB0LD`
	vampire-like, 4/3, small tail counter; 35-level residue path to HALT	**open** (100,000 levels without HALT)	$1.1 \cdot 10^{-14}$ per level → ≈ $9 \cdot 10^{13}$ levels (median $6 \cdot 10^{13}$)	≈ 5,000× the TM2 run: about 200 years on a notebook, a few years on a large cluster (simulator still to be written)	≈ $10^{(2.25 \cdot 10^{13})}$

🟡 `1RB0RE_1LC0RA`
`1LA1LD_1LC1LF`
`0LC0LB_1LE---`
machine M
	8/3 map, memory word r as a stack (table not closed); 108-step path to HALT from r = empty	**open** (probviously halting cryptid; 400 macro steps, about 10^{320} TM steps, no HALT; closest approach 102 macro steps)	about 3^{-80} per reset of r (wiki, Racheline)	out of reach	≈ $10^{10^{38}}$ (wiki, heuristic)

🟡 `1RB---_0RC0RE`
`1RD1RF_1LE0LB`
`1RC0LD_1RC1RA`
	same machine as M, start state F (in the naming of M); reaches C(12, [6]) after 111 steps	**open** (probviously halting cryptid; 400 macro steps without HALT, closest approach 103 macro steps)	as M	out of reach	≈ $10^{10^{38.1}}$ (wiki, heuristic)

🟡 `1RB1RD_1LC0LE`
`1RA0LB_1RA1RF`
`0RA0RC_1RE---`
	same machine as M, start state C; reaches C(16, empty) of M after 95 steps	**open** (400 macro steps, about 10^{320} TM steps, no HALT; closest approach 101 macro steps)	as M	out of reach	≈ $10^{10^{38}}$ (as M)

🟡 `1RB0LC_1RC1RE`
`1LA0LD_0RB0RA`
`1RB1RF_1RD---`
	same machine as M, start state B; reaches C(6, empty) of M after 18 steps	**open** (400 macro steps, about 10^{320} TM steps, no HALT; closest approach 103 macro steps)	as M	out of reach	≈ $10^{10^{38}}$ (as M)

🟡 `1RB1RF_1RC0LD`
`1LB1RA_1RE1LE`
`0RC0LB_0RB---`
machine N
	vampire-like, 4/3, nested sub-process (prefix table mod 6); table with several blocks, not closed	**open** (202 Bells, $10^{35.5}$ steps without HALT)	below $2 \cdot 10^{-6}$ per Bell (measured upper bound)	unknown	–

🟡 `1RB1LB_0RC0LD`
`1LD1RE_1RC0LA`
`1RD1RF_0RD---`
	as N (very probably equivalent)	**open**	as N	unknown	–

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